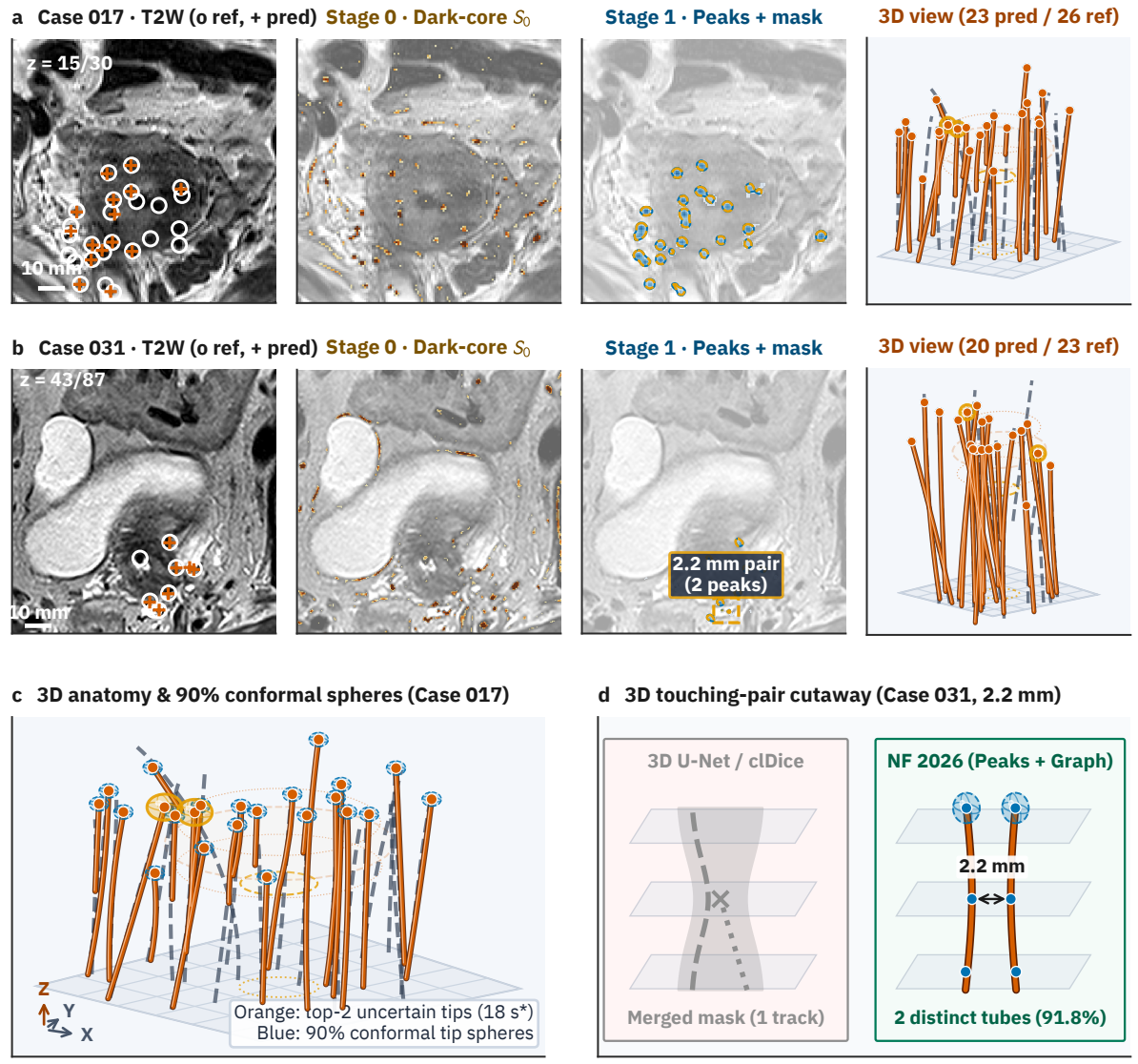

a Case 017 · T2W (o ref, + pred)
Stage 0 · Dark-core S_0
Stage 1 · Peaks + mask
3D view (23 pred / 26 ref)
z = 15/30
10 mm
b Case 031 · T2W (o ref, + pred)
Stage 0 · Dark-core S_0
Stage 1 · Peaks + mask
3D view (20 pred / 23 ref)
z = 43/87
10 mm
2.2 mm pair
(2 peaks)
c 3D anatomy & 90% conformal spheres (Case 017)
Z
Y
X
Orange: top-2 uncertain tips (18 s*)
Blue: 90% conformal tip spheres
d 3D touching-pair cutaway (Case 031, 2.2 mm)
3D U-Net / clDice
Merged mask (1 track)
NF 2026 (Peaks + Graph)
2.2 mm
2 distinct tubes (91.8%)